Engineering Ethics:

When Buildings Fail



Key Stage / Year	Subject Area	Duration
KS4 (Years 10-11)	**Design & Technology**	**50 minutes**

Lesson overview

Students learn the key facts of both disasters, focusing on how flammable exterior materials, cost-cutting, and ignored warnings contributed to each. Through a brief comparison activity, they identify the pattern of failure common to both cases. The main focus of the lesson is on engineering ethics: students engage with core principles (duty of care, honesty, accountability, whistleblowing) and write an evaluative paragraph applying these to the evidence.

Learning objectives

- Describe the key facts and timelines of the Grenfell Tower and Wang Fuk Court fires.
- Explain how material choices and structural decisions contributed to the rapid spread of fire in both cases.
- Compare the common factors across both disasters: flammable exterior materials, cost-cutting, ignored warnings, and regulatory failures.
- Evaluate the ethical responsibilities of engineers, designers, manufacturers, and regulators when safety is compromised, and apply this understanding to your own design practice.

Before teaching

Read the safeguarding guidance on page 3 and identify the school's pastoral support route. Teaching actions and resource references are on pages 4-5.

Author: Wonyong Park (University of Southampton). Reviewers: Susan Crawford and Laura Hanley.
Edition updated 17 October 2026. Artwork: The Grenfell Curriculum website.

Context and approach

Lesson context

On 14 June 2017, a fire at Grenfell Tower in North Kensington, London, killed 72 people. The fire spread catastrophically due to combustible ACM cladding fitted during a 2012-16 refurbishment. The Grenfell Tower Inquiry final report (published September 2024) found decades of failure by central government, regulators, and the construction industry, including systematic dishonesty by cladding and insulation manufacturers.

On 26 November 2025, fire engulfed seven of eight 31-storey towers at Wang Fuk Court in Tai Po, Hong Kong, killing 168 people. The buildings were undergoing facade renovation. Non-compliant scaffolding netting and expanded polystyrene (EPS) foam boards over windows contributed to external fire spread. Investigations and reporting have raised concerns about cost-cutting, falsified safety certificates and ignored resident warnings. Allegations and criminal charges must be distinguished from proven findings; dated updates are on page 9.

This lesson uses both fires as comparative case studies to explore engineering ethics: how material selection, cost-cutting, and regulatory failures can have fatal consequences, and what ethical responsibilities engineers, designers, manufacturers, and regulators bear.

EDJ connection

Start with the engineering, not the horror	The lesson focuses on material properties, design decisions, and systemic failures before addressing human impact. The engineering analysis is the learning vehicle.
Honour lived experience	Residents' warnings are presented as valid evidence that was ignored. Their knowledge of their own buildings is treated as legitimate and important.
Situate events in social and structural context	Both fires are examined as systemic failures shaped by cost-cutting, regulatory weakness, and power imbalances rather than isolated accidents.
Emphasise justice, not just memory	The ethical principles explored (duty of care, accountability, honesty) are linked to ongoing reforms and students' own future professional practice.
Support emotional safety and dialogue	Sensitivity guidance is provided. Focus is on engineering lessons, not graphic detail. Students may opt out. Teachers should identify and signpost their school's pastoral referral routes before the lesson.

Prepare the lesson

Safeguarding and trauma-informed approach

- Avoid graphic detail and distressing imagery; focus on engineering analysis, material properties, and ethical principles.
- Provide advance warning about lesson content. Allow students to step out if distressed.
- Be mindful of students who may live in tower blocks, social housing, or buildings undergoing renovation.
- Direct students to pastoral support where appropriate.

Big questions

- What ethical responsibilities do engineers and designers have when selecting materials for buildings?
- What structural and material similarities connect two fires separated by eight years and 9,600 kilometres?
- Who bears responsibility when systemic failures lead to loss of life: the manufacturer, the designer, the regulator, or the government?

Materials

- Lesson slide deck (PDF): *Engineering Ethics: When Buildings Fail*, 20 slides. Teaching guidance is in this guide; there are no speaker notes.
- Student activity sheets, printed A4 portrait: Task 1 case studies (pages 1-2), Task 2 comparison table (page 3), Task 3 Venn diagram (page 4) and Task 4 evaluative writing (page 5).
- Sticky notes or scrap paper for the plenary exit ticket; this is separate from the five worksheet pages.
- Highlighters and writing materials.

Lesson at a glance

0-5 minutes	Opening and learning objectives · Slides 1-2
5-17 minutes	Case studies and Task 1 reading · Slides 3-6
17-23 minutes	Begin comparison Tasks 2-3 · Slides 7-12
23-33 minutes	Engineering ethics discussion · Slides 13-15
33-44 minutes	Task 4 evaluative writing · Slides 16-17
44-50 minutes	Takeaways and exit ticket · Slides 18-19

Learning plan

Opening, case studies and comparison

0-5 minutes — Opening

Greet students and provide a brief sensitivity note: the lesson covers real events in which people lost their lives; the focus is on engineering analysis and ethics. Display the title slide. Ask the opening question: "What do you think an engineer's most important responsibility is?" Take 2-3 responses. Display slide 2 and read through the learning objectives.

Slides 1-2

5-17 minutes — Case studies: Grenfell Tower and Wang Fuk Court

Present both case studies in sequence, keeping each to around 6 minutes of teacher input and guided reading. For Grenfell (slides 3-4), cover: 24-storey tower; combustible cladding added during refurbishment; rapid external fire spread; 72 killed; cost-cutting on materials; warnings ignored. For Wang Fuk Court (slides 5-6), cover: eight 31-storey towers wrapped in scaffolding; non-compliant netting and foam boards over windows; rapid upward spread; 168 killed; cost-cutting on netting; residents' warnings about the foam. Do not imply that the two external wall systems were identical.

Students use Task 1 to read and highlight (a) materials, (b) fire spread and (c) failures and cost-cutting while following the presentation, without copying all details. Keep narration focused on the what and why of material failure rather than graphic detail. Check for understanding: "Can you already see a similarity between the two?"

Slides 3-6; student sheets, Task 1, pages 1-2

17-23 minutes — Comparison

Use slides 7-11 to recap the parallels briefly: flammable exterior materials, rapid upward spread, fire entering through windows, cost-cutting over safety and ignored warnings. Ask: "Which similarity do you think is the most significant and why?" Display slide 12. In pairs, students begin Task 2 (comparison table) and Task 3 (Venn diagram). Allow about 1 minute for the recap, 3 minutes to begin the table and 2 minutes to begin the Venn diagram. Finish incomplete comparison work after the lesson.

Slides 7-12; student sheets, Tasks 2-3, pages 3-4

Learning plan

Ethics, evaluative writing and reflection

23-33 minutes — Engineering ethics discussion

This is the core of the lesson. Display slide 13 and introduce five ethical principles: duty of care, honesty and transparency, accountability, professional competence, and whistleblowing. For each, give a concrete example from the case studies (e.g. manufacturers falsifying fire-safety test data as a breach of honesty; residents reporting flammable netting as attempted whistleblowing).

Facilitate a structured class discussion using slides 14-15. Key prompts: "Who bears the most responsibility - the manufacturer, the designer, the contractor, or the regulator?" "Is following the law enough, or should engineers go beyond minimum standards?" "What would you do if your employer asked you to use a material you believed was unsafe?"

Accept all reasoned answers and draw out the tension between profit, compliance, and safety. Connect to students' own D&T coursework: "When you design products, what is your duty of care?"

Slides 13-15; 2017 ethical-principles reading, page 7

33-44 minutes — Evaluative writing

Display slides 16-17. Students work individually or in pairs to complete Task 4: an evaluative paragraph of 6-8 sentences answering the key question about engineering ethics and responsibility, drawing on evidence from both case studies. Sentence starters and connectives are provided on the sheet for scaffolding. Circulate, prompt, and differentiate: support students who need it with verbal prompts; challenge stronger students to consider multiple perspectives and link to their own coursework.

Slides 16-17; student sheets, Task 4, page 5

44-50 minutes — Plenary

Display slide 18 and read through the key takeaways. Ask 2-3 students to share one sentence from their evaluative paragraph. Introduce the exit ticket on slide 19: "If you were an engineer asked to use a material you believed was unsafe, what would you do and why?" Give students 2 minutes to write a response on a sticky note or scrap paper. Collect as formative assessment. End with a calm reflection and signpost pastoral support.

Slides 18-19; sticky notes or scrap paper

Glossary and extension

Engineering ethics	The moral principles that govern the professional conduct of engineers, including responsibility for public safety, honesty in testing, and accountability for design decisions.
Cladding	Material applied to the exterior of a building, typically for insulation and weatherproofing. ACM cladding has aluminium sheets sandwiching a core that may be combustible (e.g. polyethylene).
Compartmentation	A fire safety strategy that divides a building into enclosed sections (compartments) to prevent fire from spreading between them.
Chimney effect	The tendency of hot air and fire to rise rapidly through a vertical channel, drawing oxygen from below and accelerating flames. It is one possible mechanism of upward spread; the two fires did not have identical external wall systems.
Duty of care	A legal and ethical obligation to ensure that products and buildings do not pose an unreasonable risk of harm to users or occupants.
Systemic failure	A failure caused by multiple interconnected problems within a system (regulation, industry, management) rather than a single isolated error.
Whistleblowing	Reporting unsafe or unethical practices to an authority or the public, often at personal risk, in order to prevent harm.

Keywords: Engineering ethics, cladding, flammability, chimney effect, compartmentation, facade, duty of care, systemic failure, cost-cutting, regulatory failure

Extension activities or homework

- **Option 1 (15-20 minutes):** Research the Lakanal House fire (2009) or the Lacrosse Building fire, Melbourne (2014). Write 200-300 words connecting it to the lessons from the Grenfell and Wang Fuk Court case studies.
- **Option 2 (15 minutes):** Write a design specification for a hypothetical building renovation project that prioritises fire safety. Reference at least three lessons learned from the two case studies.
- **Option 3 (10 minutes):** Read the Engineering Council and Royal Academy of Engineering *Statement of Ethical Principles* (2017). Identify three principles and explain how each was violated at Grenfell or may apply to the reported failures at Wang Fuk Court. Distinguish confirmed findings from allegations.

Background reading

For teachers before the lesson

Grenfell Tower Inquiry - Phase 2 Report: Executive Summary (September 2024)

The official final report of the public inquiry. Covers decades of government, regulatory and industry failure, and the "systematic dishonesty" of cladding and insulation manufacturers.

Read the official report

"How the products used in Grenfell Tower's cladding system were tested and sold", Inside Housing (2020)

Detailed investigative journalism on the ACM panels and Celotex insulation - how internal warnings were suppressed, fire tests were manipulated, and cheaper combustible products were marketed for high-rise use.

Read the Inside Housing article

"Tai Po fire timeline: Wang Fuk Court's HK$330 million renovation, ignored warnings and aftermath", Hong Kong Free Press (17 December 2025)

Independent journalism tracing the renovation project from 2016 through to the fire, including alleged bid-rigging, resident complaints to multiple government departments, and the arrests of contractor and consultancy directors. Read as a dated account; see the updates on page 9.

Read the HKFP timeline

"Hong Kong's Wang Fuk Court fire and the global lessons for high-rise safety", International Fire & Safety Journal (5 December 2025)

Technical analysis drawing on NFPA's Fire & Life Safety Ecosystem framework. Compares the systemic failures at Wang Fuk Court with those at Grenfell and explains how multiple weaknesses aligned catastrophically.

Read the technical analysis

Statement of Ethical Principles (2017 revision), Royal Academy of Engineering & Engineering Council

The four-principle framework retained for this lesson: honesty and integrity; respect for life, law, the environment and public good; accuracy and rigour; leadership and communication. Short, accessible, and directly relevant to a UK engineering or design curriculum. The five classroom themes on slide 13 are teaching prompts, not a verbatim list of this statement's headings.

2017 statement (legacy document link)

Professional Code of Conduct for Designers (2020, revised 2022), International Council of Design (ICoD)

The international standard for professional design ethics. Covers honesty, integrity, responsibility to humanity, advocacy and education, and the duty to consider the social, cultural, environmental and safety impacts of design decisions throughout a product's life cycle.

Read the professional code

Curriculum connections

Design & Technology (KS4)

Selected GCSE specification connections. These are curriculum links for the case studies, not claims that building fire safety is a separately prescribed topic.

AQA 8552	3.1.1: new and emerging technologies, including ethical and environmental evaluation. 3.1.6: materials and working properties. 3.3.1: investigation using primary and secondary data. 3.3.2: environmental, social and economic challenge. These support materials analysis, ethical selection and evaluation of evidence.
OCR J310	Section 2c, Principles of Design and Technology: 3.1 and 3.3 (wider impacts, sustainability and social/ethical awareness); 5.2 (material selection, including properties, cost and ethical considerations). Safe design and responsible selection connect the cases to these principles.
Pearson Edexcel 1DT0	1.13: selecting materials and processes for the context. 1.14: environmental, social and economic challenges, including material cost. Component 2, Investigate: investigation of needs and research, including existing products and materials. These support comparison of evidence and evaluation of design choices.

Cross-curricular connections

- **Science:** material properties, combustion, polymers, thermal conductivity.
- **Citizenship:** social responsibility, governance, the role of public inquiries, accountability.
- **Geography:** urbanisation, housing inequality, the built environment.

Specification sources

- AQA 8552 subject content
- OCR J310 specification (version 2.0, 2026)
- Pearson Edexcel 1DT0 specification (issue 4, November 2022)

Acknowledgements

Author: Wonyong Park (University of Southampton). Reviewers: Susan Crawford and Laura Hanley.

Project and illustrations: The Grenfell Curriculum. Website artwork is used selectively with project attribution; no illustrator name was supplied. The uncredited incident photographs in the original deck are not reproduced.

Edition updated 17 October 2026. Original teaching files are preserved in the editable source package.

Evidence and updates

Teacher reference · factual updates dated to their sources

Grenfell: materials and cost

Use specific insulation terms: mainly Celotex RS5000 polyisocyanurate (PIR), with some Kingspan K15 phenolic insulation. The polyethylene core of the ACM panels was the principal fuel for external spread; a cavity-only "chimney effect" account is insufficient.

The £293,368 figure was a proposed saving for aluminium in place of zinc cladding with cassette fixing. It is not a verified price difference between combustible and fire-resistant ACM cores. The pack retains the figure with that context.

Inquiry Phase 2, volume 2 · volume 7 · 28 July 2020 transcript

Wang Fuk Court: figures and dated status

The Hong Kong Police confirmed 168 deaths, including one firefighter, on 15 January 2026. The source pack's "16 arrested on charges" is replaced: by 10 June 2026, 35 people had been arrested and seven individuals and two companies had been charged with 25 offences. On 9 September 2026, the Buildings Department announced 259 additional prosecutions involving two companies and four people. Charges and allegations are not findings of guilt.

Reported netting prices of HK$54 and HK$100 per roll, applied to 2,300 rolls, imply a simple price difference of HK$105,800 (about HK$106,000). This is an illustrative calculation, not an audited total saving. It replaces the inconsistent HK$115,000 figure. Other reported details retained from the original case summary are labelled as reporting, not final inquiry findings.

Police identification update, 15 January 2026 · Police/ICAC charges, 10 June 2026 · Buildings Department, 9 September 2026

Teaching the comparison responsibly

Compare the shared ethical issues without asserting an identical fire mechanism. "Chimney effect" remains in the glossary, but distinguish Grenfell's external wall system from Wang Fuk Court's temporary renovation materials. Describe reports about windows, alarms, warnings and certificates as reported concerns where findings remain unsettled.

The 2017 ethical-principles statement is deliberately retained. The lesson's five classroom themes remain unchanged. The legacy document link on page 7 may redirect; obtain the 2017 edition when using that reading, rather than substituting a newer framework.